

DIMUTHU WIJESIRIWARDENE

ROBOTICS & AUTONOMOUS SYSTEMS RESEARCHER

MOBILE ROBOTICS • PERCEPTION & CONTROL

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Location : Colombo, Sri Lanka

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| ● Application Engineer | Synopsys, Inc. | Jun 2022 - Present |
| ● Graduate Research Assistant | Faculty of Engineering
University of Sri Jayewardenepura
Colombo, Sri Lanka | Oct 2021 - Jun 2022 |
| ● Associate Software Engineer | Insharp Technologies Pvt. Ltd.
Colombo, Sri Lanka | Aug 2020 - Oct 2021 |
| ● Undergraduate research assistant (Internship) | Sustainable Computing Research Group(SCoRe Lab)
University of Colombo School of Computing
Colombo, Sri Lanka | Nov 2017 - Jan 2018 |

PROFESSIONAL BACKGROUNDACADEMIC BACKGROUND

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| ● Bachelor's degree <ul style="list-style-type: none">o BEng (Hons) Electrical and Electronic Engineeringo Upper Second Class Honours | Curtin University
Perth, Australia | Jan 2016 - Feb 2020 |
| ● GCE Advanced Level <ul style="list-style-type: none">o Physical Science Streamo Math, Physics & Chemistry | Dharmapala Vidyalaya Pannipitiya
Colombo, Sri Lanka | Jan 2011 - Sep 2014 |

PROJECTS

- NovasPlay - Omnidirectional Soccer robot ([Demo Video](#))
 - o Built a remote-controlled omnidirectional soccer playing robot featuring a custom dribbling mechanism
 - o Modeled the robot in Autodesk Fusion, fabricated it using laser-cut sheet metal and 3D-printed components, assembling them into a functional robot
 - o Achieved a 5-1 victory in the initial competition round - Competition project at Synopsys (2024)
- Fire rescue assisting robot
 - o Designed a thermal vision based system for human detection in a smoke filled environments using low cost sensor mounted on a UGV/UAV
 - o Final year individual research project at Curtin University (2019)
- Reverse Vending Machine ([Demo Video](#))
 - o Designed and simulated a 3D model of a reverse vending machine in Blender and built a working prototype
 - o A research project at the University of Sri Jayewardenepura in collaboration with industry partner OREL Corporation (Pvt.) Ltd. (2022)
- Low-cost reflectometer
 - o Designed and built a low-cost time-domain reflectometer to locate faults in cables as part of a human-elephant conflict mitigation system - Research project at SCoRe Lab (2017)
- Fence energizer reverse engineering
 - o Reverse engineered the GALLAGHER B180 Multi Power Fence Energizer to build a low-cost energizer as a part of a research project at SCoRe Lab (2017)
- 3D printed quadruped robot
 - o Built a four-legged robot as a part of a research project at SCoRe Lab (2018)

- Verdi - Debug and Verification Management Platform
 - Performed incoming bug analysis, root cause analysis, and validation of new features and enhancements for customers such as Apple, Intel, NVIDIA, AMD, etc.
 - Developed test automation solutions using AI-Driven testing methodologies and scripting; contributed new regression tests to the regression suite, improving verification coverage and quality assurance processes
 - PV owner of Verdi Transaction Debug component with quality assurance and signoff responsibilities
 - Worked with Synopsys VIP designs & UVM, performing tool specific testing - Project at Synopsys
- VCS - Verilog Compiled Simulator
 - Validated and debugged VCS Intelligent Coverage Optimization (ICO), an AI/ML driven technology for accelerating functional coverage closure in semiconductor verification
 - Validated the integration of ICO with Verdi RDA (Regression Debug Automation), enabling customers to identify faulty code and traces more effectively - Project at Synopsys
- Euclide - Real-Time Hardware Development and Verification IDE
 - Validation and verification of new features and enhancements of the RTL tool
 - Analyzed customer bugs and identified root causes - Project at Synopsys
- jTerm - Efficient Jira creation tool
 - Developed a tool to create Jiras without workflow interruptions, with built-in support for screenshots and annotations via the in-house tool *Snipper*, and globally utilized with 500+ Jiras created to date
 - Presented at the internal poster presentation event ("Purple Poster Event", 2025) at Synopsys
- jCrash - Streamlined crash reporting tool
 - Built a tool to report application crashes instantly by executing a single command
 - Awarded third place in the 2024 Innovations Award at Synopsys
 - Presented at the internal poster presentation event ("Purple Poster Event", 2024) at Synopsys
- gCron - Crobjob manager
 - Built an intuitive interface for cron job management with natural-language scheduling, real-time previews and backup support; adopted by teams globally
 - Presented at the internal poster presentation event ("Purple Poster Event", 2026) at Synopsys
- weGen - All-in-one weekly report generator
 - Built a unified interface to update project statuses, manage tasks and generate weekly progress reports
 - A Project at Synopsys (2025)
- PerfTest - An automated script for performance testing with multiple customer designs
 - Recognised for streamlining the process of manual performance testing - Project at Synopsys (2024)
- Cdiff - Customised diff command
 - Recognised in the Innovations Awards - Project at Synopsys (2023)
- TuTi - Virtual Classroom
 - Full stack development - Project at Insharp Technologies (2021)
- Retailz - Data analytics tool for supermarkets and suppliers
 - Full stack development - Project at Insharp Technologies (2020)
- The Wind Turbine ([Demo Video](#))
 - Presenter and team member
 - Project for Principles and Communication module at the university (2016)
- The Automated Bedsheet ([Demo Video](#))
 - Engineered a solution from initial idea through conceptual design to a functional prototype as project lead
 - Researched, designed and fabricated custom stepper motor drivers from scratch, reducing the cost by 85%
 - Presented at the university engineering exhibition ("Young Engineering Expo", 2016)

TECHNICAL SKILLS

- Programming & Software
 - Python, C, Shell scripting (Csh, Bash), Tcl
 - Version control (Git)
 - Linux (Ubuntu, Raspbian)
 - AI-assisted development
 - Prompt engineering
- CAD, Design & Fabrication
 - Autodesk Fusion, Blender, AutoCAD, SolidWorks
 - 3D printing (PLA, ABS, TPU, TPE), sheet metal fabrication (laser cutting)
 - Mechanical assembly; workshop tools (drills, saws, sanders, compressors)
- Computer Vision, Simulation & Numerical Tools
 - OpenCV, pandas, NumPy, SciPy
 - MATLAB
 - Godot Engine
- Validation, Verification & Quality Assurance
 - Synopsys VCS, Verdi (Simulation, Debugging)
 - Root Cause Analysis, Jira
 - Selenium, Test Planning
 - AI-based test case creation agents
- Embedded Systems & Robotics
 - Raspberry Pi, Arduino, PIC, MSP430, PLC
 - Sensors, actuators
 - Robot Operating System (ROS 2)
 - Hardware interfacing
 - I2C, SPI, UART, PWM
- Electronics & PCB Design
 - Circuit simulation: NI Multisim, Proteus
 - PCB design: Eagle PCB
 - PCB manufacturing: toner-transfer method
 - PCB reverse engineering
 - Oscilloscopes, Multimeter, Soldering
- Web & Software Development
 - Django, Laravel
 - HTML, CSS, JavaScript, Vue.js
 - Blockchain (Smart contracts, Solidity)
- Technical Communication & Media
 - Poster presentation
 - Technical demonstrations
 - Video production
 - Video editing (Adobe Premiere Pro)

Awards, Presentations & Recognitions

- 2026 - Presenter, Purple Poster Event (gCron)
- 2025 - Presenter, Purple Poster Event (jTerm)
- 2024 - 3rd Place, Synopsys Innovations Award (jCrash)
- 2024 - Presenter, Purple Poster Event (jCrash)
- 2024 - Competitor, Synopsys RoboCup Competition (NovasPlay - [Demo Video](#))
- 2016 - Presenter, Young Engineering Expo (Automated Bedsheet - [Demo Video](#))

Publications

- Dimuthu Wijesiriwardene and Rohana Thilakumara. "Human Detection for Fire Rescue Robots Using a Thermal Sensor and Hu Moment Invariants.", Manuscript submitted to *IEEE International Conference on Electrical Engineering and Emerging Technologies (ICEET)*, 2026.

References

Dr. Udaya Wijenayake
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